Question 1

Create a simple greeting script similar to HelloHackersArise

Approach

I created a basic bash script that prints a greeting message when executed.

Commands

- `nano` → create script file
- `echo` → display message
- `chmod +x` → make script executable

Output

STEP 1 : create script file

```
┌──(aman㉿kali)-[~]
└─$ nano greeting.sh
```

STEP 2 : add script code

```
  GNU nano 8.7.1
#!/bin/bash
echo "Hello Hacker, welcome to bash scripting"
```

STEP 3 : make executable

```
┌──(aman㉿kali)-[~]
└─$ chmod +x greeting.sh
```

STEP 4 : run script

```
┌──(aman㉿kali)-[~]
└─$ ./greeting.sh
Hello Hacker, welcome to bash scripting
```

Key Observations

Script executes commands automatically when run.

What I learned

Bash scripts automate repeated tasks.

Question 2

Create script to scan systems for MSSQL service on port 1433

Approach

I created a script that scans IP addresses to check if port 1433 (Microsoft SQL Server is open.)

Commands

- `for` → loop through IP addresses
- `nc` → check open port
- port 1433 → MSSQL default port

Output

STEP 1 : create script

```
┌──(aman㉿kali)-[~]
└─$ nano MSSQLscanner.sh
```

STEP 2 : script code

```
  GNU nano 8.7.1
#!/bin/bash

for ip in 192.168.1{1..10}
do
nc -zv $ip 1433 2>/dev/null && echo "$ip MSSQL port open"
done
```

STEP 3 : make executable

```
┌──(aman㉿kali)-[~]
└─$ chmod +x MSSQLscanner.sh
```

STEP 4 : run script

```
┌──(aman㉿kali)-[~]
└─$ ./MSSQLscanner.sh
```

Key Observations

Script checks multiple systems automatically.

What I learned

Port scanning can be automated using bash loops.

Question 3

Modify scanner to accept IP range and port from user input

Approach

I modifies the script so user enters starting IP, ending IP, and port number. Script displays only open ports.

Commands

- `read` → take user input
- `seq` → generate number range
- `nc` → test port connectivity

Output

STEP 1 : edit script

```
┌──(aman㉿kali)-[~]
└─$ nano MSSQLscanner.sh
```

STEP 2 : updated script

```
GNU nano 8.7.1
#!/bin/bash

echo "Enter starting IP last octet: "
read start

echo "Enter ending IP last octet: "
read end

echo "Enter port number: "
read port

for i in $(seq $start $end)
do
ip="192.168.1.$i"
done
```

STEP 3 : run script

```
┌──(aman㉿kali)-[~]
└─$ ./MSSQLscanner.sh
Enter starting IP last octet:
1
Enter ending IP last octet:
50
Enter port number:
443
```

Key Observations

Script dynamically scans based on user input.

What I learned

Scripts can interact with users and adapt behaviour dynamically.

Security Insights

1. Scripts can automate repetitive hacking or admin tasks.
2. Port scanning helps identify exposed services like databases.
3. User input in scripts can be exploited if not validated.